

Phase 3 — Infrastructure Engineering

Goal

Understand how production infrastructure works, learn cloud computing, Linux administration, networking, infrastructure automation, containers, Kubernetes, monitoring, and deployment while extending your Phase 1 and Phase 2 backend systems into production environments.

Topic 1 — Virtualization & Cloud Fundamentals

To Learn: Virtual Machines, Hypervisors, Type 1 vs Type 2 Hypervisors, Hardware Virtualization, Containers vs VMs, Cloud Computing Models, IaaS, PaaS, SaaS, Regions, Availability Zones, Data Centers, Shared Responsibility Model
Books: Cloud Computing: Concepts, Technology & Architecture
Resources: AWS Cloud Practitioner Docs, Google Cloud Architecture Center
YouTube: NetworkChuck, IBM Technology, freeCodeCamp
Blogs: AWS Cloud Concepts, Google Cloud Architecture Center, Cloudflare Learning Center
Strategy: Understand how physical servers become cloud infrastructure
Projects: Create and manage local virtual machines using VirtualBox

Topic 2 — Linux System Administration

To Learn: Linux Filesystem, Boot Process, Kernel, Processes, Threads, Signals, Users, Groups, Permissions, File Ownership, Package Management, Systemd, Services, Cron Jobs, Environment Variables, Logs, SSH, Disk Management
Books: How Linux Works, The Linux Command Line
Resources: Linux Documentation, Arch Wiki
YouTube: freeCodeCamp, Learn Linux TV, The Linux Foundation
Blogs: Linux Documentation, Arch Linux Wiki, Red Hat Linux Administration Documentation
Strategy: Manage Linux servers daily instead of only using commands
Projects: Setup and manage a production-like Linux server

Topic 3 — Linux Networking

To Learn: Network Interfaces, IP Addressing, Subnetting, Routing Tables, TCP/IP Stack, Ports, Sockets, Firewalls, iptables, nftables, DNS Configuration, Network Troubleshooting, netstat, ss, tcpdump
Books: TCP/IP Illustrated Vol.1
Resources: Linux Networking Documentation, Cloudflare Learning Center
YouTube: Practical Networking, Hussein Nasser, freeCodeCamp
Blogs: Cloudflare Learning Center, Linux Network Documentation, Red Hat Networking Guide
Strategy: Debug every network problem using Linux tools
Projects: Build Network Monitoring CLI

Topic 4 — Shell Scripting & Automation

To Learn: Bash Basics, Variables, Conditions, Loops, Functions, Arguments, Exit Codes, Pipes, Redirection, Text Processing, awk, sed, grep, Automation Scripts
Books: Bash Cookbook
Resources: Bash Reference Manual
YouTube: freeCodeCamp, Traversy Media, NetworkChuck
Blogs: Bash Reference Manual, ShellCheck Documentation, Linux Command Documentation
Strategy: Automate every repeated infrastructure task
Projects: Server Backup Script, Log Analyzer, Deployment Script

Topic 5 — AWS Fundamentals

To Learn: AWS Global Infrastructure, Regions, Availability Zones, IAM Basics, Compute Services, Storage Services, Networking Services, Security Model, Pricing Concepts
Books: AWS Certified Solutions Architect Study Guide
Resources: AWS Documentation, AWS Well Architected Framework
YouTube: freeCodeCamp, Stephane Maarek, AWS Events
Blogs: AWS Documentation, AWS Well Architected Framework, AWS Architecture Center
Strategy: Map every infrastructure concept to AWS services
Projects: Deploy a simple application on AWS

Topic 6 — AWS Compute (EC2)

To Learn: EC2 Instances, AMI, Instance Types, EBS Volumes, Snapshots, Security Groups, SSH Access, User Data, Elastic IP, Auto Recovery
Books: AWS Certified Solutions Architect Associate
Resources: AWS EC2 Documentation
YouTube: AWS EC2 Full Course, Stephane Maarek, Adrian Cantrill
Blogs: AWS EC2 Documentation, AWS EC2 Best Practices, AWS Compute Blog
Strategy: Replace local servers with cloud servers
Projects: Deploy Node.js Backend on EC2

Topic 7 — AWS Identity & Security

To Learn: IAM Users, Roles, Policies, Permissions, Least Privilege, Access Keys, MFA, Security Groups, Network ACLs, Secrets Management
Books: AWS Security Best Practices
Resources: AWS IAM Documentation
YouTube: AWS IAM Full Course, Stephane Maarek, NetworkChuck
Blogs: AWS IAM Documentation, AWS Security Best Practices, AWS Security Blog
Strategy: Design infrastructure with security first
Projects: Secure Multi-Service AWS Environment

Topic 8 — AWS Networking (VPC)

To Learn: VPC, CIDR, Subnets, Public Subnets, Private Subnets, Route Tables, Internet Gateway, NAT Gateway, VPC Peering, Security Groups
Books: AWS Networking Cookbook
Resources: AWS VPC Documentation
YouTube: AWS VPC Full Course, Adrian Cantrill, freeCodeCamp
Blogs: AWS VPC Documentation, AWS Networking Blog, VPC Design Best Practices
Strategy: Design cloud networks before deploying applications
Projects: Production VPC Architecture

Topic 9 — Cloud Storage

To Learn: Object Storage, Block Storage, File Storage, S3 Buckets, Storage Classes, Lifecycle Rules, Versioning, Encryption, EBS, EFS
Books: AWS Storage Best Practices
Resources: AWS S3 Documentation
YouTube: AWS S3 Full Course, Stephane Maarek, AWS Storage Deep Dive
Blogs: AWS S3 Documentation, AWS Storage Blog, AWS Storage Best Practices
Strategy: Choose storage based on workload requirements
Projects: File Storage Service using S3

Topic 10 — Infrastructure as Code (Terraform)

To Learn: Infrastructure as Code, Terraform Architecture, Providers, Resources, Variables, State Files, Modules, Remote Backend, Terraform Cloud, Workspaces
Books: Terraform Up & Running
Resources: Terraform Documentation
YouTube: freeCodeCamp, HashiCorp Terraform Tutorials, TechWorld with Nana
Blogs: Terraform Documentation, HashiCorp Learn Terraform, Terraform Registry
Strategy: Never manually create production infrastructure
Projects: Build Complete AWS Infrastructure using Terraform

Topic 11 — Configuration Management (Ansible)

To Learn: Configuration Management, Inventory, Playbooks, Tasks, Variables, Handlers, Roles, Templates, Vault, Idempotency
Books: Ansible for DevOps
Resources: Ansible Documentation
YouTube: TechWorld with Nana, NetworkChuck, Red Hat Ansible Training
Blogs: Ansible Documentation, Red Hat Ansible Blog, Ansible Galaxy Documentation
Strategy: Automate server configuration
Projects: Automated Server Setup

Topic 12 — Docker Internals

To Learn: Docker Architecture, Images, Containers, Layers, Dockerfile, Volumes, Networks, Registry, Container Runtime, Namespaces, cgroups

Books: Docker Deep Dive

Resources: Docker Documentation

YouTube: TechWorld with Nana, Nigel Poulton, freeCodeCamp

Blogs: Docker Documentation, Docker Blog, Container Runtime Documentation

Strategy: Understand what happens behind docker run

Projects: Containerize Backend Application

Topic 13 — Kubernetes Fundamentals

To Learn: Kubernetes Architecture, Control Plane, Nodes, Pods, Deployments, ReplicaSets, Services, Namespaces, ConfigMaps, Secrets, Labels, Selectors

Books: Kubernetes Up & Running

Resources: Kubernetes Documentation

YouTube: TechWorld with Nana, freeCodeCamp, KubeCon Talks

Blogs: Kubernetes Documentation, Kubernetes Blog, CNCF Learning Resources

Strategy: Deploy everything inside Kubernetes

Projects: Deploy Backend Application Cluster

Topic 14 — Kubernetes Advanced

To Learn: StatefulSets, DaemonSets, Jobs, CronJobs, Persistent Volumes, Storage Classes, Ingress, Network Policies, RBAC, Operators

Books: Kubernetes Patterns

Resources: Kubernetes Documentation

YouTube: Kubernetes Production Best Practices, KubeCon Advanced Sessions, TechWorld with Nana

Blogs: Kubernetes Documentation, CNCF Blog, Kubernetes Patterns

Strategy: Learn production Kubernetes patterns

Projects: Production Kubernetes Deployment

Topic 15 — Helm & Kubernetes Packaging

To Learn: Helm Charts, Templates, Values, Releases, Repositories, Dependency Management

Books: Kubernetes in Action

Resources: Helm Documentation

YouTube: Helm Kubernetes Tutorial, TechWorld with Nana, Kubernetes Official Tutorials

Blogs: Helm Documentation, Artifact Hub Documentation, Kubernetes Helm Charts Guide

Strategy: Package applications like production teams

Projects: Create Helm Charts for Backend Stack

Topic 16 — CI/CD Infrastructure

To Learn: Build Pipelines, Deployment Pipelines, GitHub Actions, Jenkins, Artifact Management, Automated Testing, Blue-Green Deployment, Canary Deployment

Books: Continuous Delivery

Resources: GitHub Actions Docs

YouTube: GitHub Actions Full Course, Jenkins Pipeline Tutorial, Continuous Delivery Explained

Blogs: GitHub Actions Documentation, Jenkins Documentation, Martin Fowler Continuous Delivery

Strategy: Automate every deployment step

Projects: Complete CI/CD Pipeline

Topic 17 — Monitoring & Observability

To Learn: Metrics, Logs, Traces, Monitoring Architecture, Prometheus, Grafana, Alerting, SLIs, SLOs, SLAs

Books: Site Reliability Engineering

Resources: Prometheus Docs, Grafana Docs

YouTube: Grafana Labs Tutorials, Prometheus Full Course, TechWorld with Nana Monitoring Stack

Blogs: Prometheus Documentation, Grafana Documentation, OpenTelemetry Documentation

Strategy: If you cannot measure it, you cannot operate it

Projects: Application Monitoring Dashboard

Topic 18 — Logging & Debugging

To Learn: Centralized Logging, Log Aggregation, ELK Stack, Loki, Log Levels, Structured Logging, Distributed Debugging

Books: Observability Engineering

Resources: Elastic Documentation

YouTube: ELK Stack Full Course, Loki Logging Tutorial, Distributed Logging Explained

Blogs: Elastic Documentation, Grafana Loki Documentation, OpenTelemetry Logging Documentation

Strategy: Debug production failures using logs

Projects: Centralized Logging System

Topic 19 — Infrastructure Security

To Learn: Server Hardening, SSH Security, Secrets Management, Encryption, TLS

Certificates, Vulnerability Scanning, Container Security, Cloud Security

Books: Practical Cloud Security

Resources: OWASP, AWS Security Center

YouTube: AWS Security Full Course, Practical DevSecOps Tutorials, NetworkChuck Cloud Security

Blogs: OWASP Documentation, AWS Security Center, Cloud Security Alliance

Strategy: Secure every layer of infrastructure

Projects: Harden Production Infrastructure

Final Project

Production Cloud Platform

Build:

- Node.js Backend
- PostgreSQL Database
- Redis Cache
- Docker Containers
- Kubernetes Deployment
- AWS Infrastructure
- Terraform Automation
- Ansible Configuration
- CI/CD Pipeline
- Monitoring Stack
- Centralized Logging